Milestone One Guidelines and Rubric

Module B: AI for Leaders

Milestone One: Identifying a Problem and Relevant Datasets

In preparation for this assignment, you will research possible fields of interest and problems currently faced by those fields. You will also work to identify currently available datasets relevant to your fields of interest. You will select **at least five datasets.**

Note: Milestone One is a fully individual assignment. Your individual analysis of the datasets will help identify the overall value of the datasets. This analysis can be completed asynchronously and during live sessions.

In this assignment, you will submit a 5- to 8-page Microsoft Word document (excluding the title page and appendix) with double spacing, 12-point Times New Roman font, and one-inch margins. **You must use the required font sizes.** Adjusting the font size, spacing, or margins down so that you can reach the required page count will be noted by the teaching team during grading and you will be penalized. Sources should be cited according to APA style. Minor differences in citation style (for example, using a slightly older version of APA) will not result in a deduction of points. The primary purpose of the citation requirement is to ensure that the grader can easily locate the sources you are referencing, if needed. No more than one page should be used to display examples of the datasets.

Criteria for Milestone One include:

- A description or summary of the nature and content of each dataset
    - What is it about?
    - What kinds of data does it include?
    - It is preferable that the datasets each contain as many types of data as possible (numbers, strings, images, audio, etc.).



- For each dataset, choose 1-2 learning types that may be applicable for analysis. Make sure you include at least two of the following within the **entire paper**:
    - Supervised and/or unsupervised learning
    - Clustering
    - Regression
    - Classification
    - Anomaly detection
- **Note:** the number of tabular, image or NLP datasets is up to you, but we recommend that you **mostly focus on tabular data** for now. **We discourage the use of time series data as they will not be selected next semester.**
- An explanation of missing, incorrect, or duplicated data
    - At least one of the datasets should contain some missing, incorrect, or duplicated data so that you can demonstrate your ability to deal with these problems.
- An explanation of the kinds of questions that could be answered by each dataset
    - Why are these questions interesting to you?
    - Why or how are these questions important for a particular industry?
        - Try to include two or three different industries across your five datasets.
        - Include an explanation about why the given industries are interesting to you personally.
- An attached weekly Capstone Development Reflection Journal
    - An attached file containing your weekly Capstone Development Reflection Journal is provided.
    - A minimum **of five weeks'** worth of Capstone Development Reflection Journal entries should be completed.
    - Note that the Capstone Development Reflection Journal entries are not part of the main Milestone submission and therefore do not count towards the 5-8 pages of the Milestone. You can re-use some material from the Journal entries to help in writing Milestone One, but if so, you must recognize and restructure it enough that the Milestone document makes sense, flows naturally, and addresses the needed questions.
- A summary of your experiences with the project to this point. Consider including the following:
    - Process: How did your work develop up to this point?
    - Learning or successes: What new skills or knowledge did you gain through this project work? What is one way you can apply your skills or knowledge when you begin to work with your group in the next phase of the project? What do you think was your biggest area of success so far in this project?
    - Challenges: Did you encounter any challenges or problems? If so, how did you address the challenges or

problems? What was the most difficult part of the project so far?
    - Only one summary is required per Milestone, whereas the reflection journal will have one entry per week.
- While your dataset may include images as part of the data, you are not required to include any figures or graphs representing your analysis in your submission for Milestone One.
- An appendix which describes any use of AI in writing this document.
    - Please include any prompts that you used as well as their output, as well as a short description of how you used AI. You may include a link to your conversation with AI. This appendix does not count toward the maximum length of your paper, and you cannot assume that the grader will read it in detail.
    - If you did not use AI, please still include an appendix stating that you did not use it.

Submission Outline
Your score can be reduced by up to 10 points out of 100 for not following this rubric and using these exact section labels.

Main body:
1. Description or summary of the nature and content of each dataset.
2. Explanation of missing, incorrect, or duplicated data.
3. Evaluation and explanation of how and why each dataset is suitable for each of several machine learning methods.
4. Industry and problem: explanation of the kinds of questions that could be answered by each dataset.
5. Summary of your experiences with the project to this point.

Required appendix (does not count toward total page count):
6. Citations
7. Reflection: weekly journal - attached weekly Capstone Development Reflection Journal.
8. AI appendix

Milestone One Rubric

Rubrics are graded on specific criteria. You will receive a grade from the LF based on where your assignment falls in the following scale:

- Exemplary: Meets all or nearly all expectations. 100% of points in the criteria
- Proficient: While meeting most expectations, the submission is lacking some details or insights. It is otherwise mostly correct and well-written. 85% of points in the criteria
- Needs improvement: While meeting many expectations and being correct in many ways, the submission is overly brief, incomplete, superficial, inaccurate, unclear and/or minimal in some ways. 70% of points in the criteria
- Not evident: Does not discuss and/or incorrect. 0% of points in the criteria

The table below shows what you need to do to achieve an "Exemplary" for each category.

Criteria	Exemplary	Points
Description of each dataset	Provides a thorough, accurate, and detailed description of five datasets, including nature, content, and types of data. **The five datasets do not need to be connected in any way.**	20
Evaluation of datasets	Provides a comprehensive and detailed evaluation of five datasets including suitability of learning type (i.e supervised or unsupervised), clustering, anomaly detection, regression, and classification.	20
Data quality	The datasets chosen are of sufficient quality to support robust analysis	5
Industry/problem	Provides a detailed or insightful explanation of the kinds of questions each dataset can answer, including what questions are important for	10

	specific industries.	
Reflection: Weekly Journal	Provides detailed weekly progress in a journal document with at least five weekly capstone assignments	5
Reflection Summary	Provides a comprehensive and insightful reflection summary, highlighting key learning experiences and overall project impact	20
Communication	Consistently and effectively communicates ideas clearly, concisely, and effectively for a specific audience	10
Citations	Incorporates all necessary citations/sources in proper APA format	5
AI Appendix	Includes an AI appendix clearly stating use of AI in submission.	5
Total		**100**